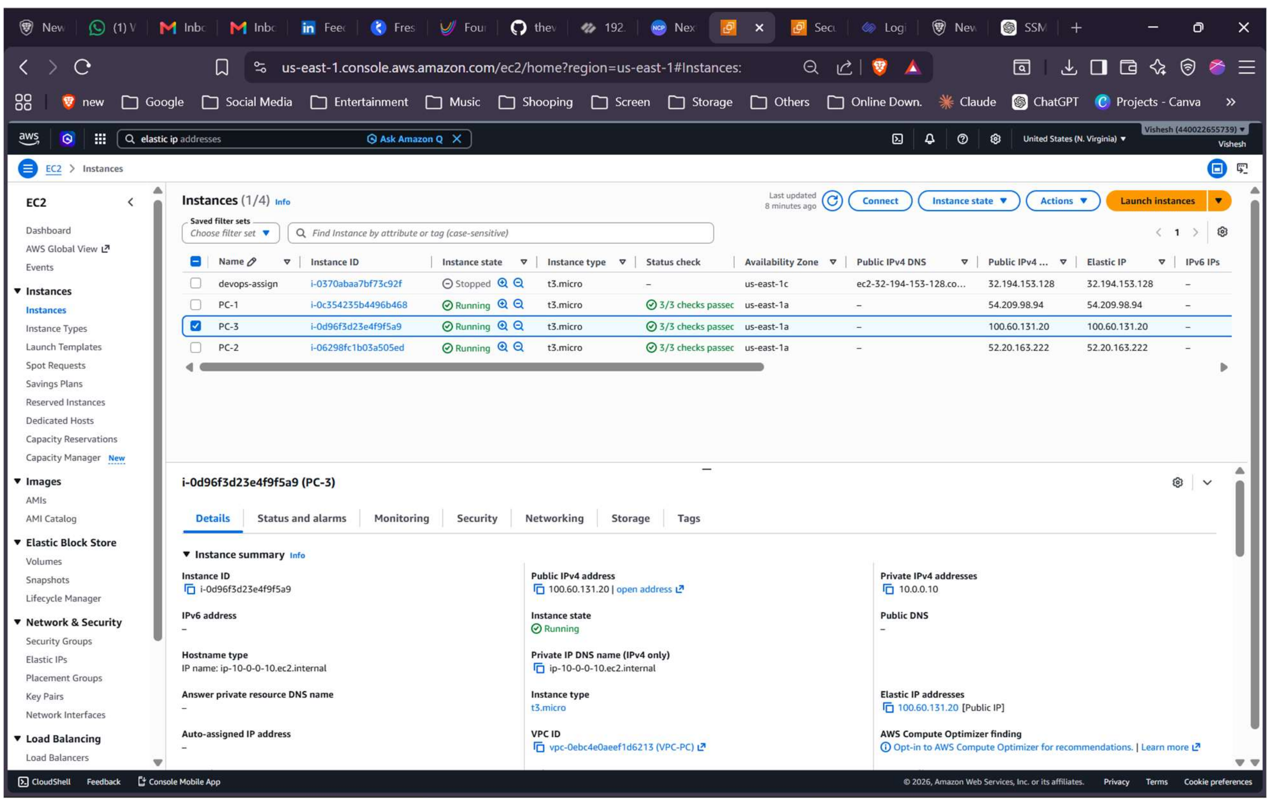
AWS Active Directory Domain Join
&
OU Management Lab

Objective:

Configure a Windows client to communicate with an Active Directory Domain Controller, join the client to the lab.local domain, rename the computer, move the computer object to the correct Organizational Unit (OU), and verify successful domain communication.

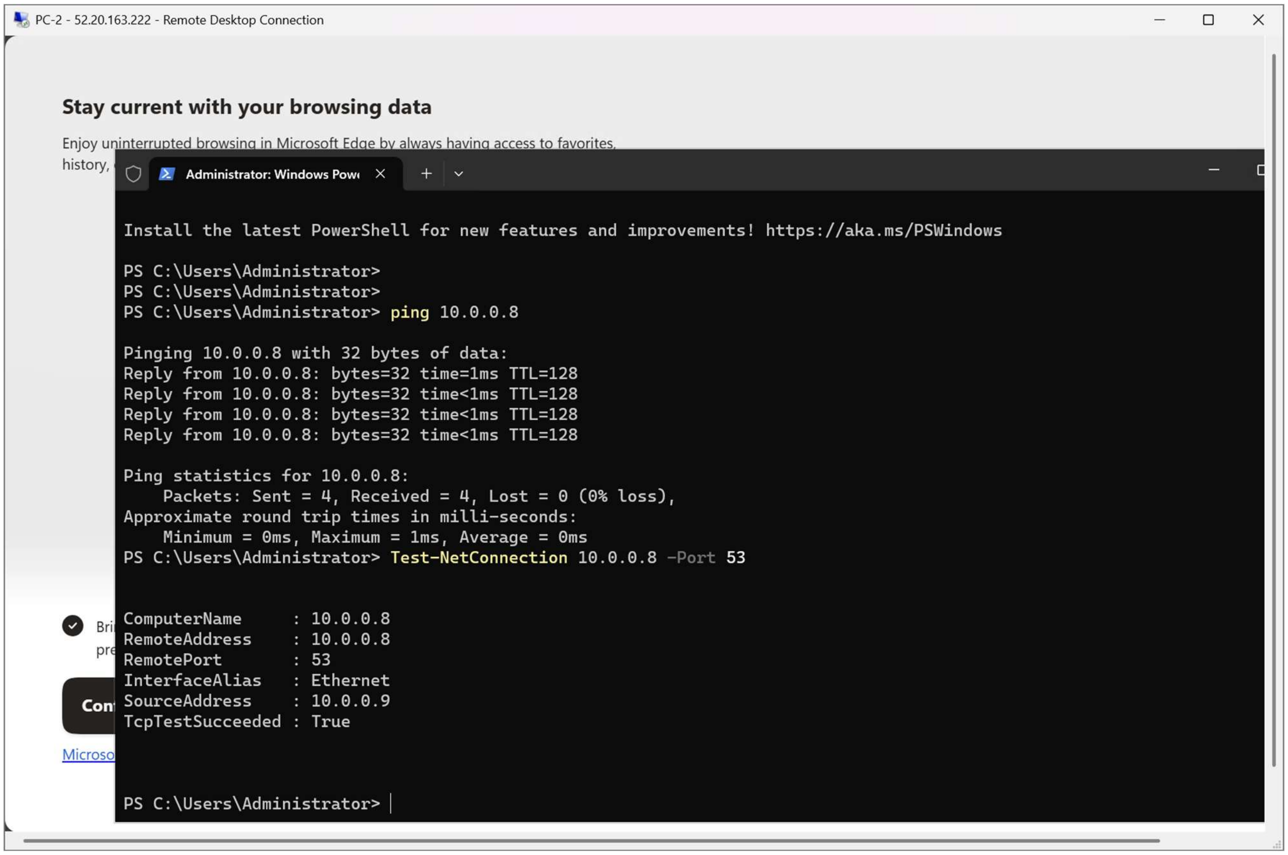
Step 1 – Verify AWS Lab Environment

Verify that the required EC2 instances were running. PC-1 hosts the Active Directory Domain Controller, PC-2 is the client machine that will join the domain, and PC-3 is available for future use.



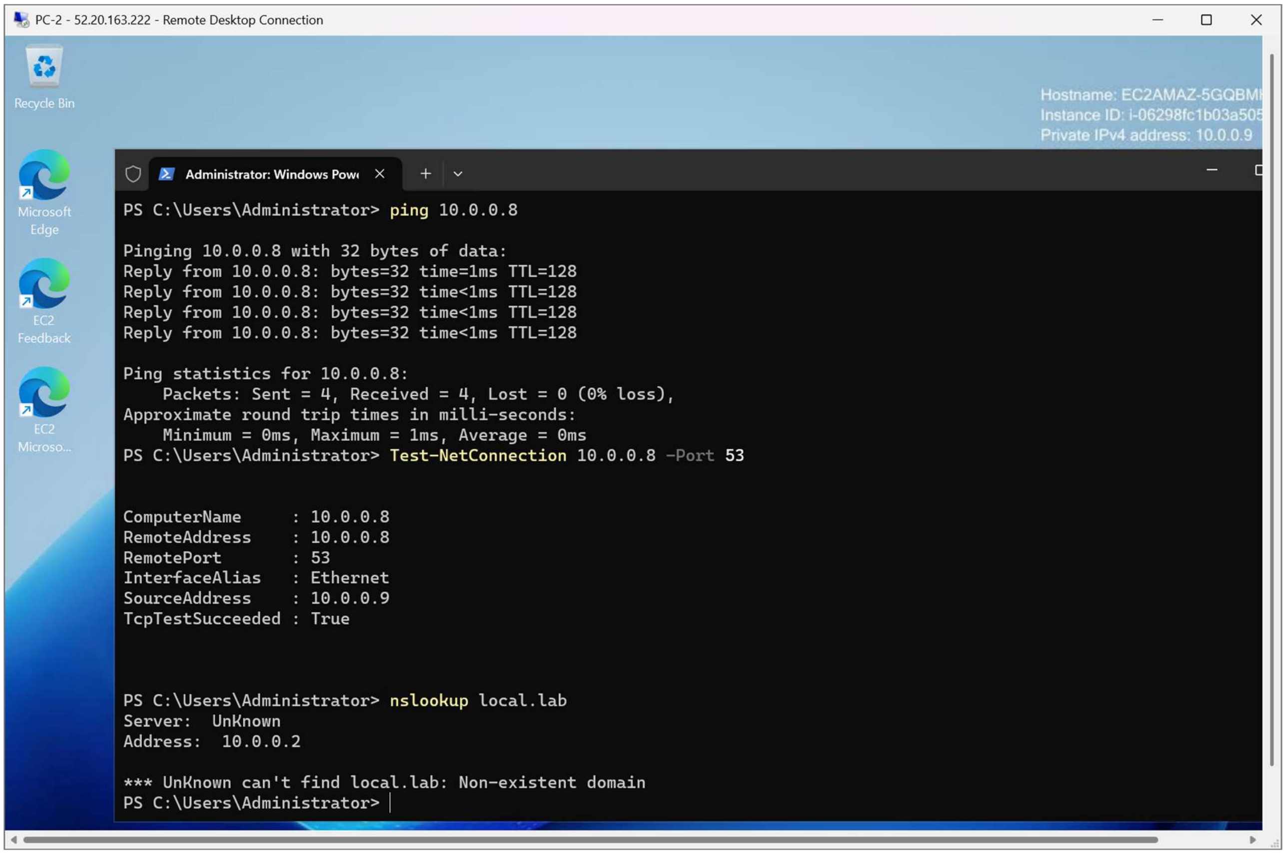
Step 2 – Test Basic Network Connectivity

From PC-2, verified Layer 3 connectivity to the Domain Controller (10.0.0.8) using ping.
Then verified that TCP port 53 (DNS) was reachable using Test-NetConnection.



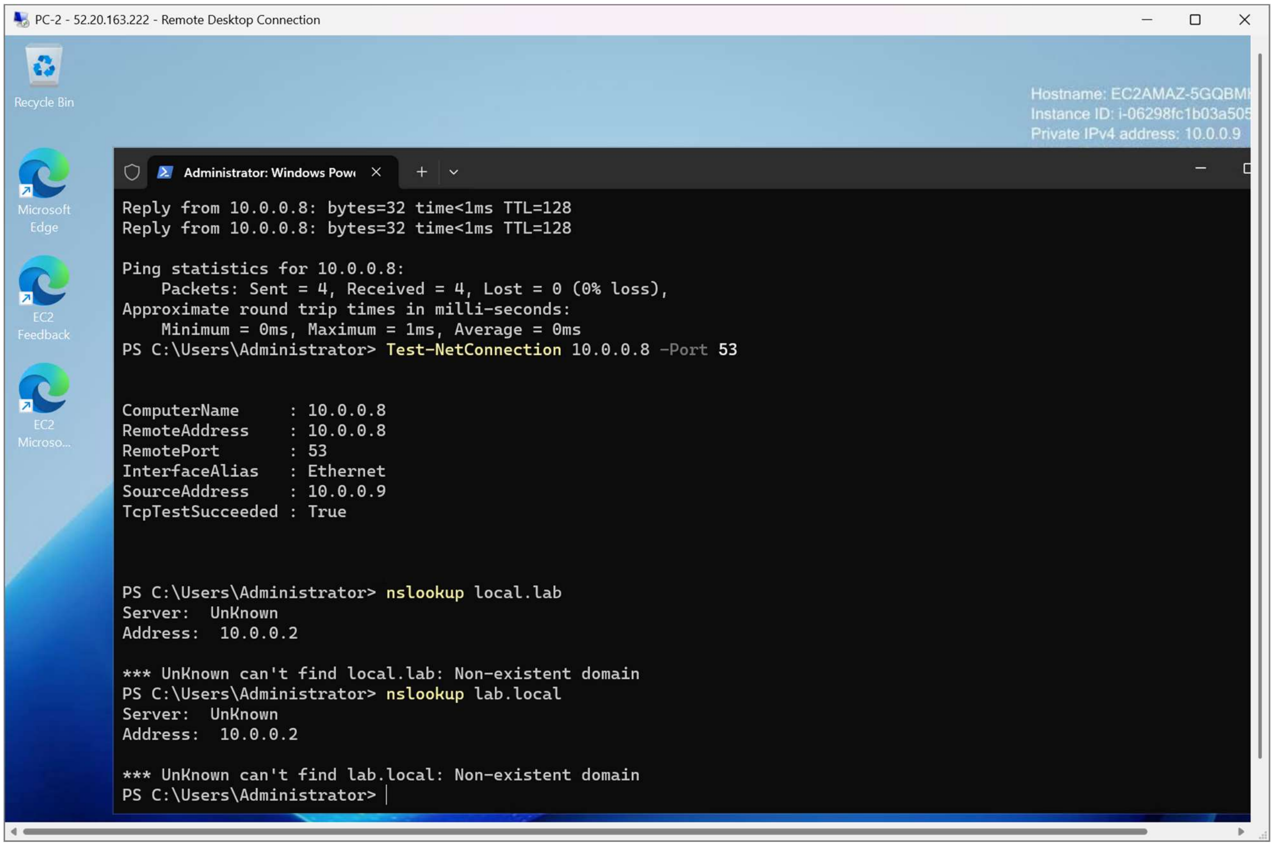
Step 3 – Identify the DNS Resolution Problem

Attempt to resolve the domain using nslookup. The query failed with 'Non-existent domain', confirming that the client was not using the correct DNS server.



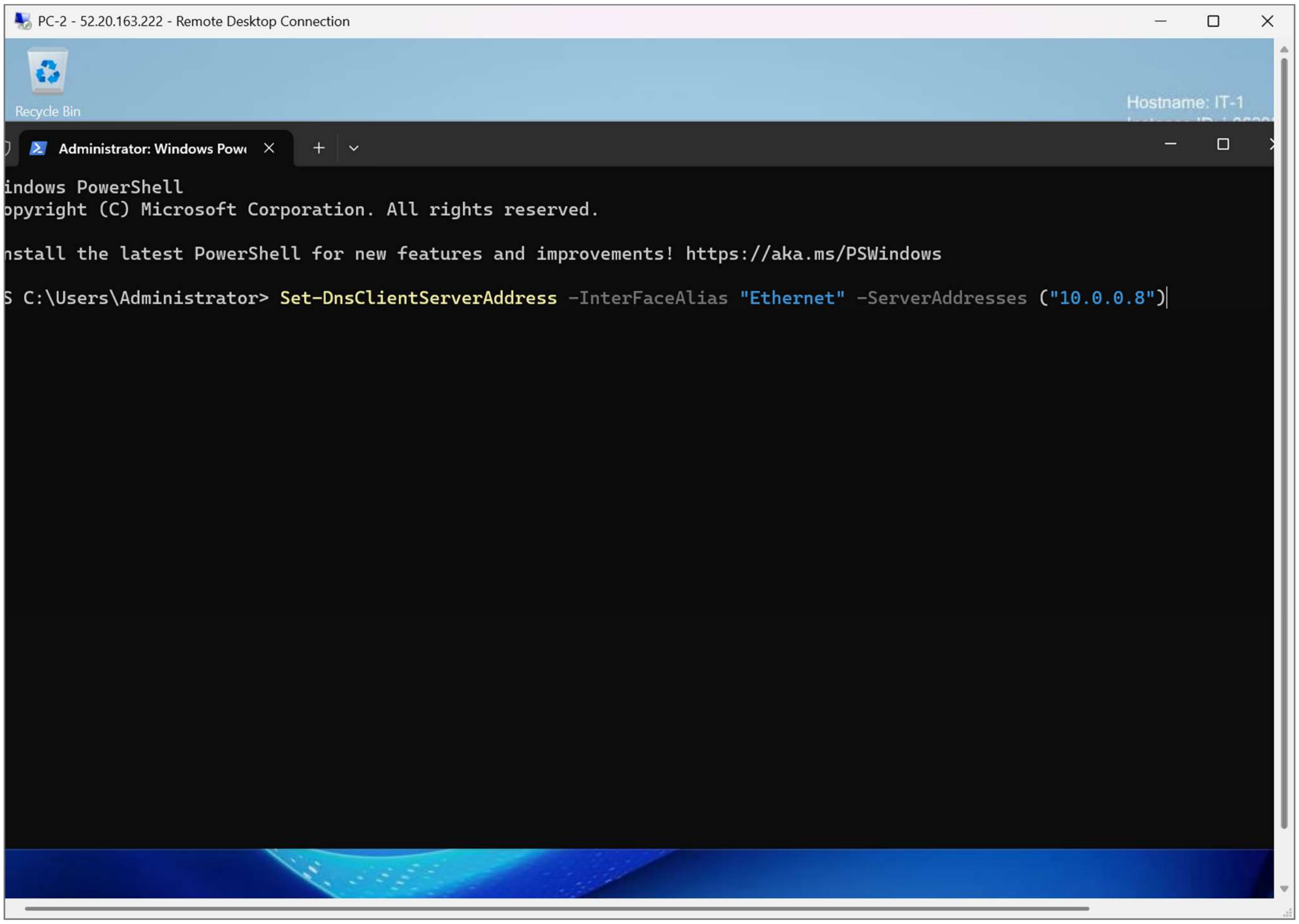
Step 4 – Verify Client Network Configuration

Execute ipconfig /all to inspect the client's IP configuration. Confirmed that the DNS server was incorrectly configured as 10.0.0.2 instead of the Domain Controller.



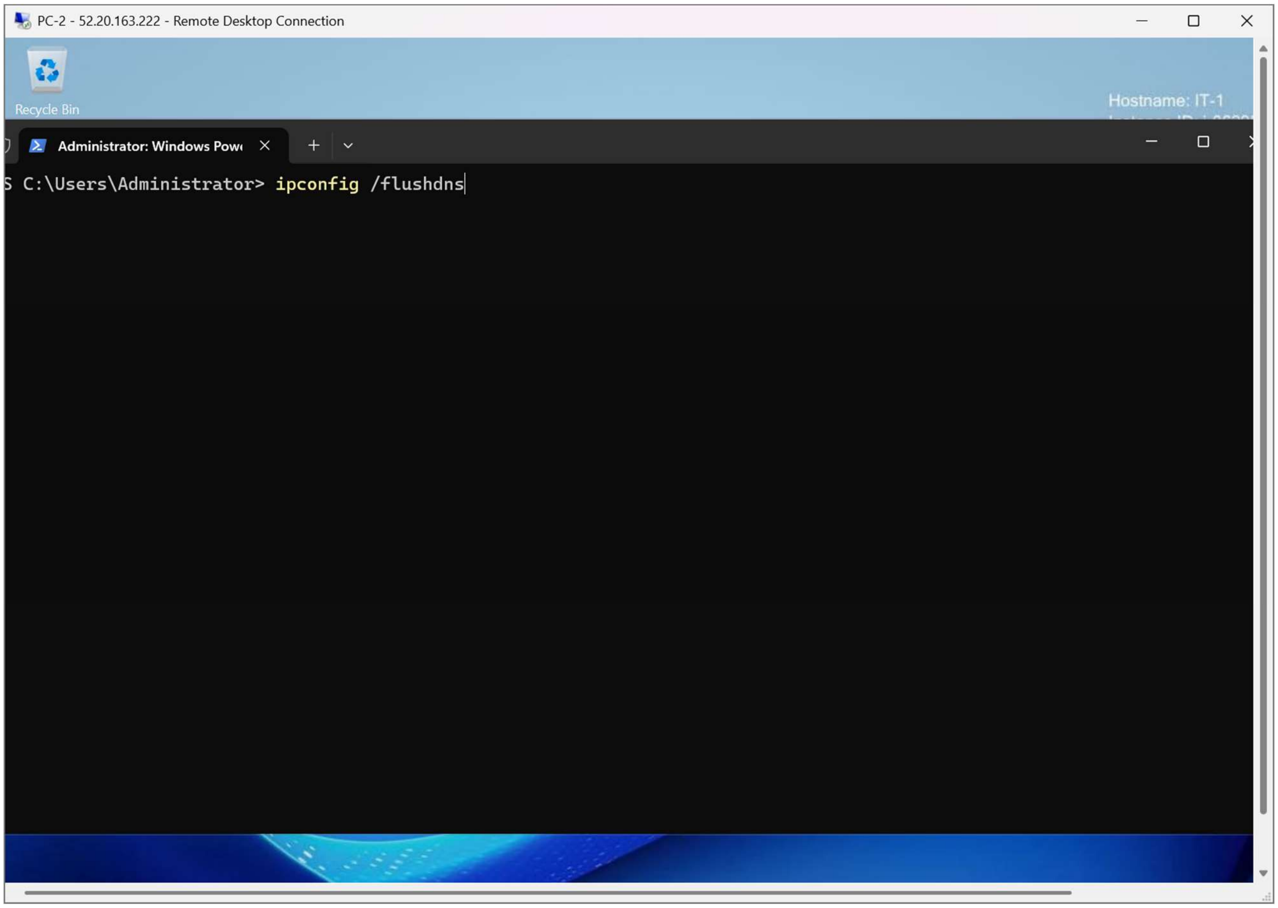
Step 5 – Configure the Correct DNS Server

Change the Preferred DNS Server on the client to the Domain Controller IP address (10.0.0.8).



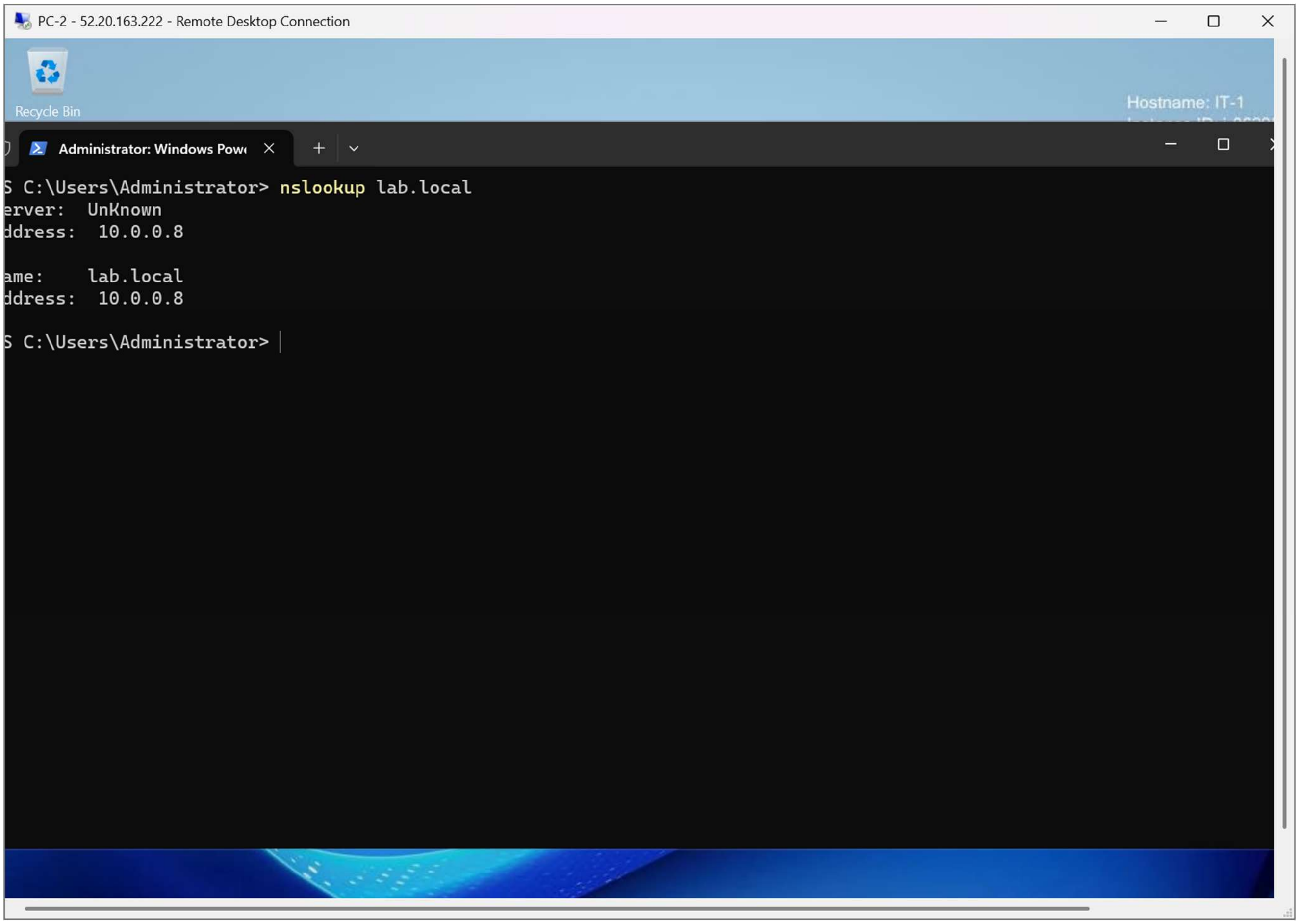
Step 6 – Clear Cached DNS Records

Execute ipconfig /flushdns to remove cached DNS entries so that future queries use the updated DNS configuration.



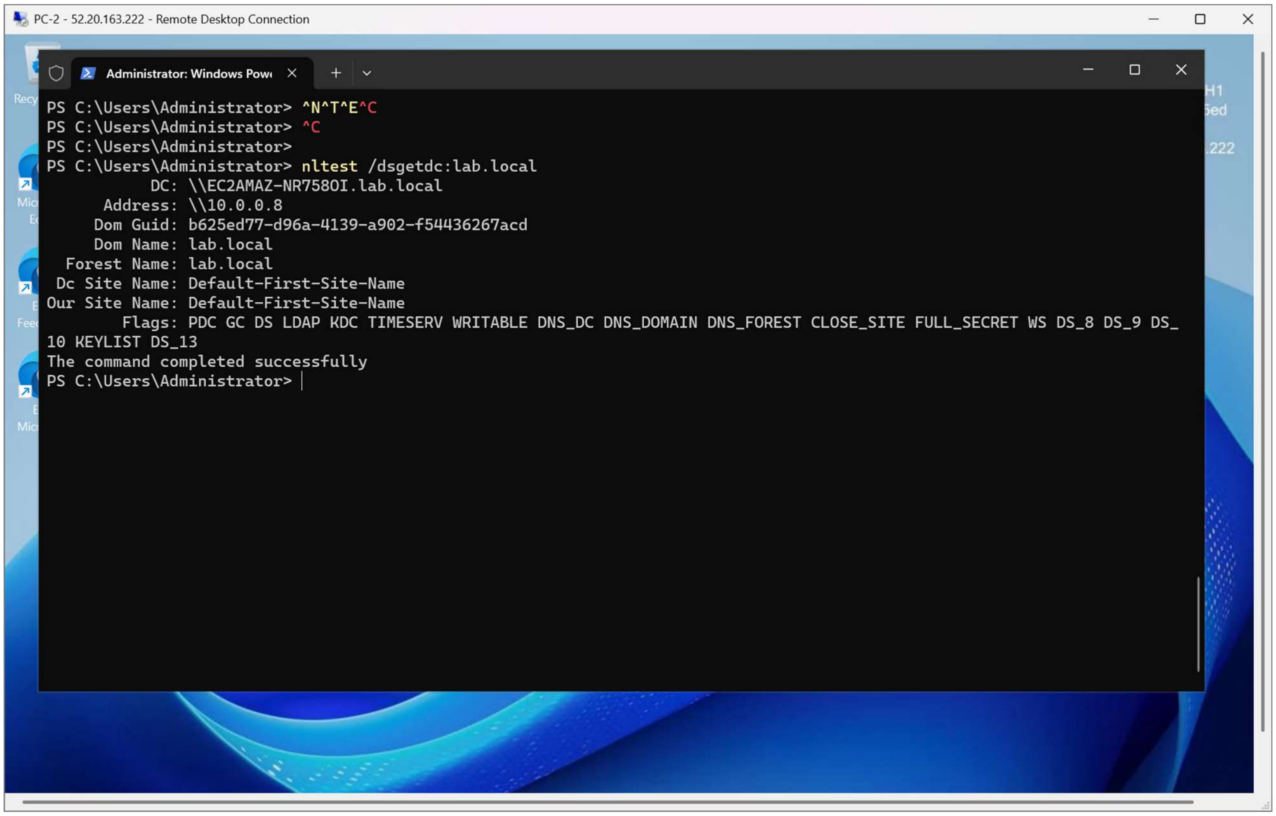
Step 7 – Verify DNS Resolution

Run nslookup lab.local again. The query now resolved successfully to 10.0.0.8, confirming that DNS was functioning correctly.



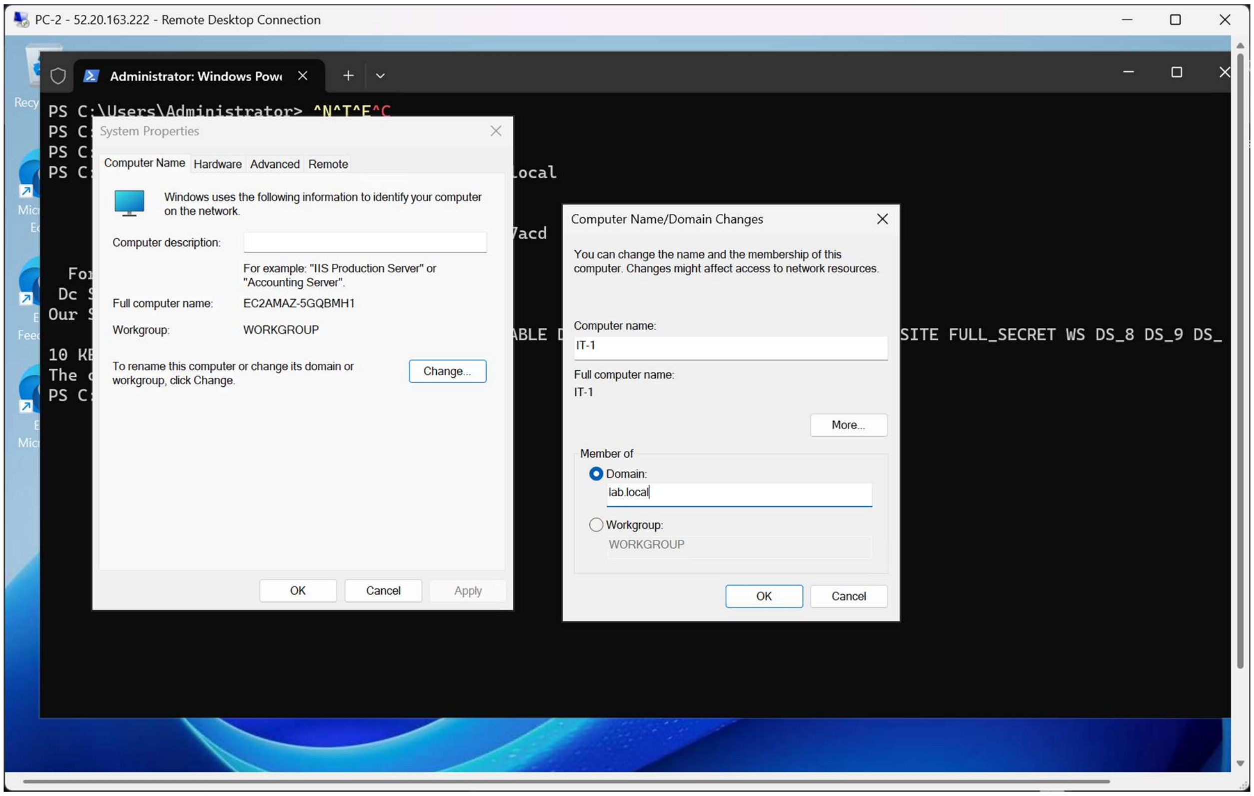
Step 8 – Verify Domain Controller Discovery

Execute nltest /dsgetdc:lab.local to verify that the client could locate the Active Directory Domain Controller.



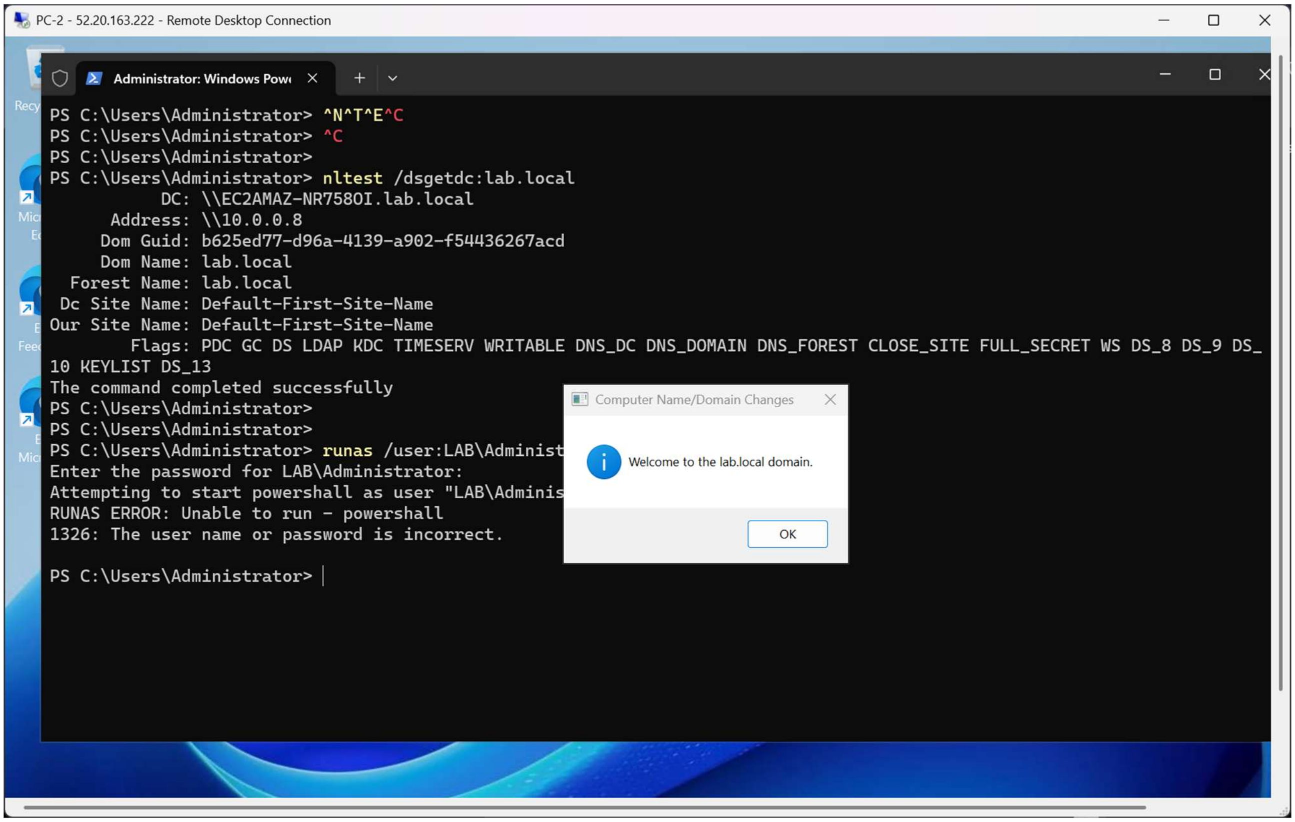
Step 9 – Join the Client to the Domain

Open System Properties, changed the computer membership from Workgroup to the lab.local domain, provided domain administrator credentials.



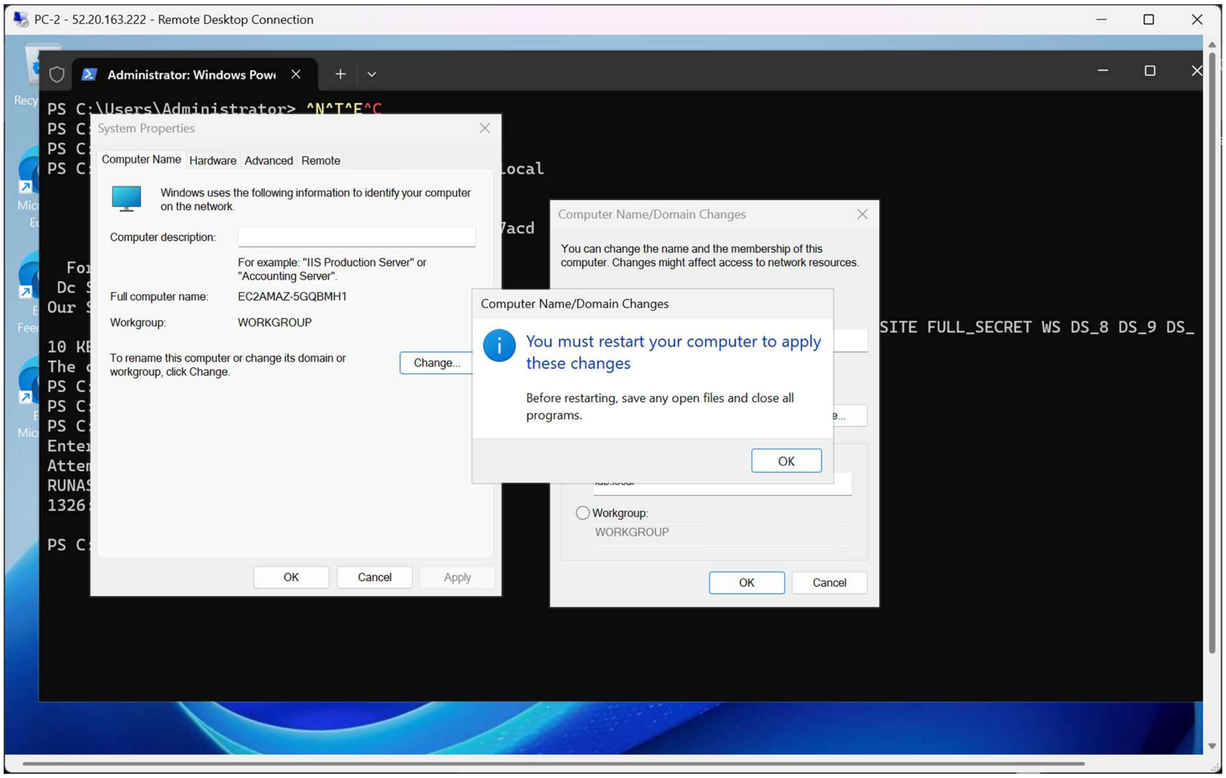
Step 10 – Rename the Client Computer

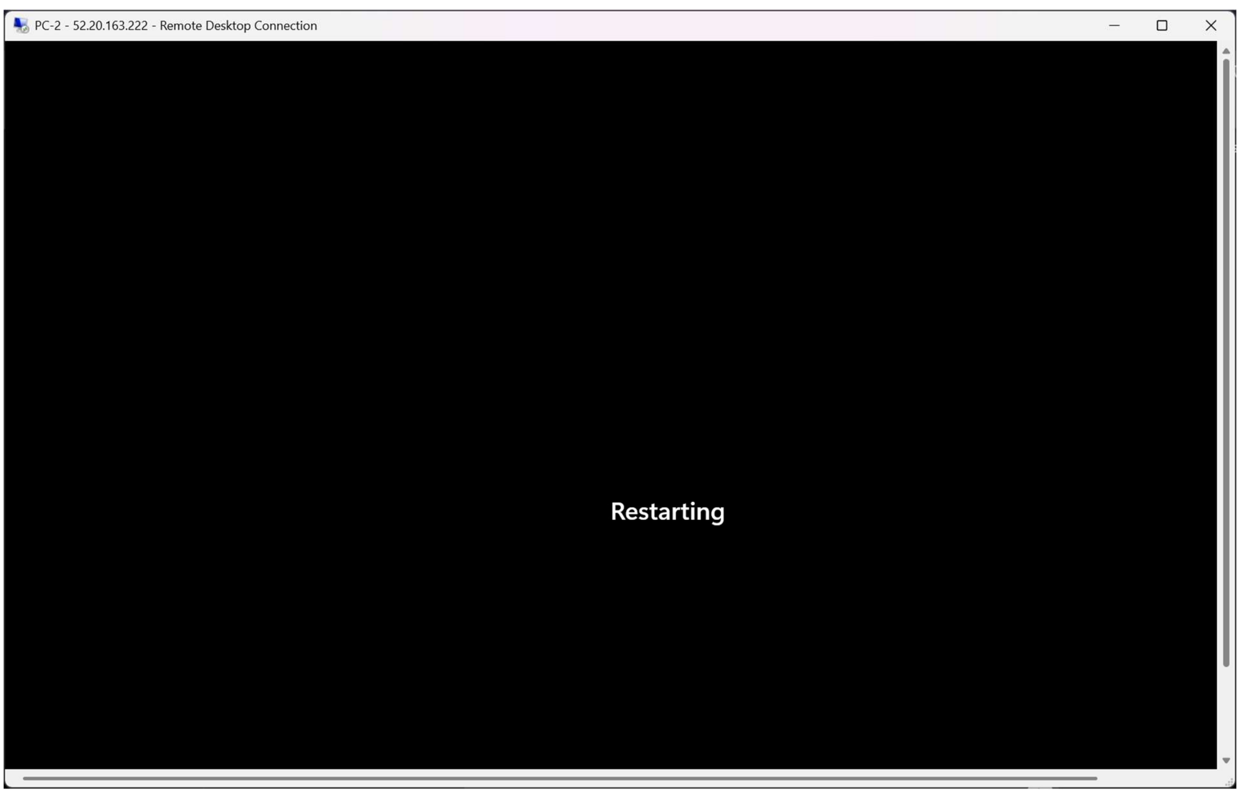
Change the default computer name to IT-1 while keeping it joined to the lab.local domain, ,
and received the 'Welcome to the lab.local domain' message



Step 11 – Restart the Client

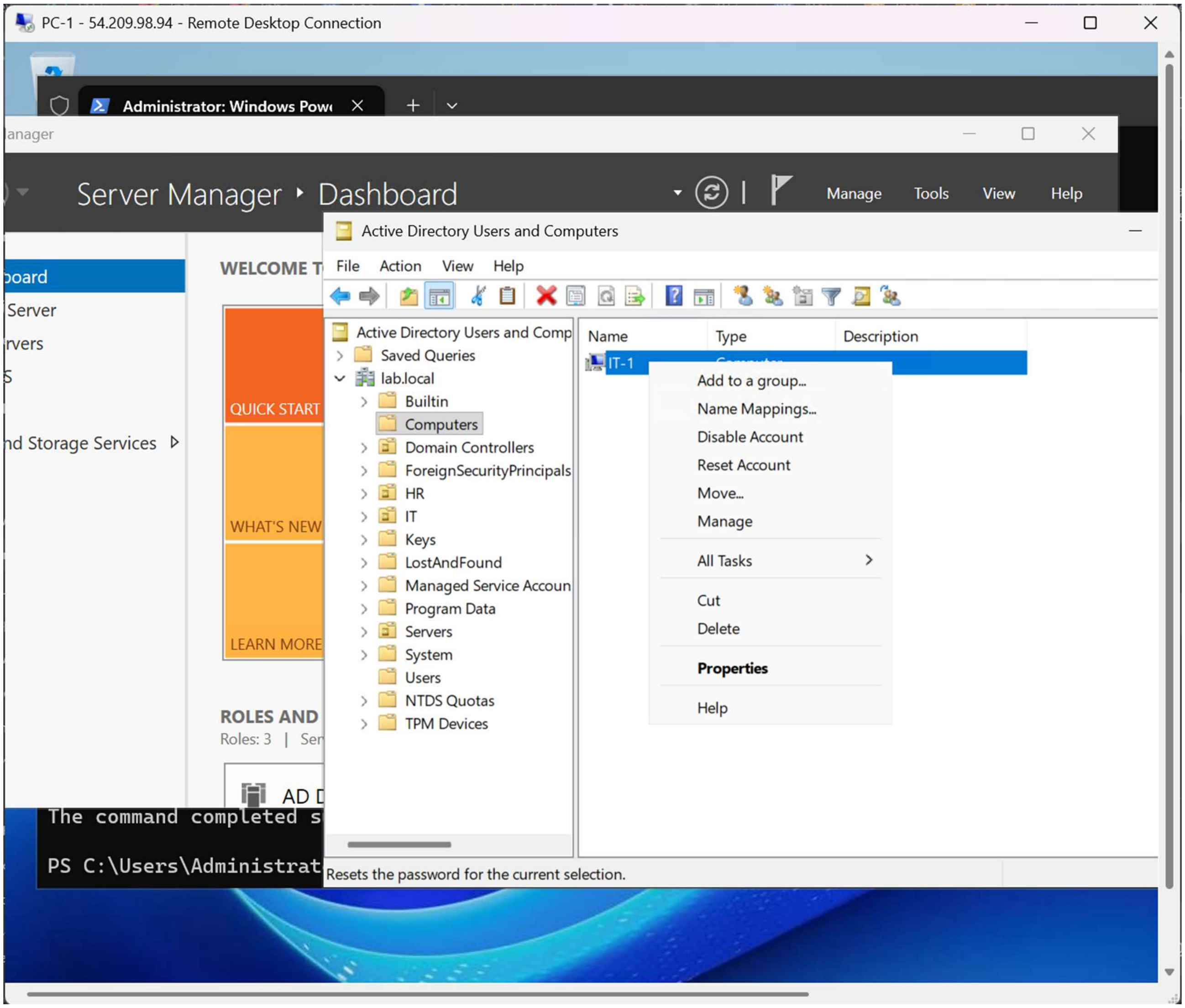
Restart the client computer to apply both the domain membership and hostname changes.





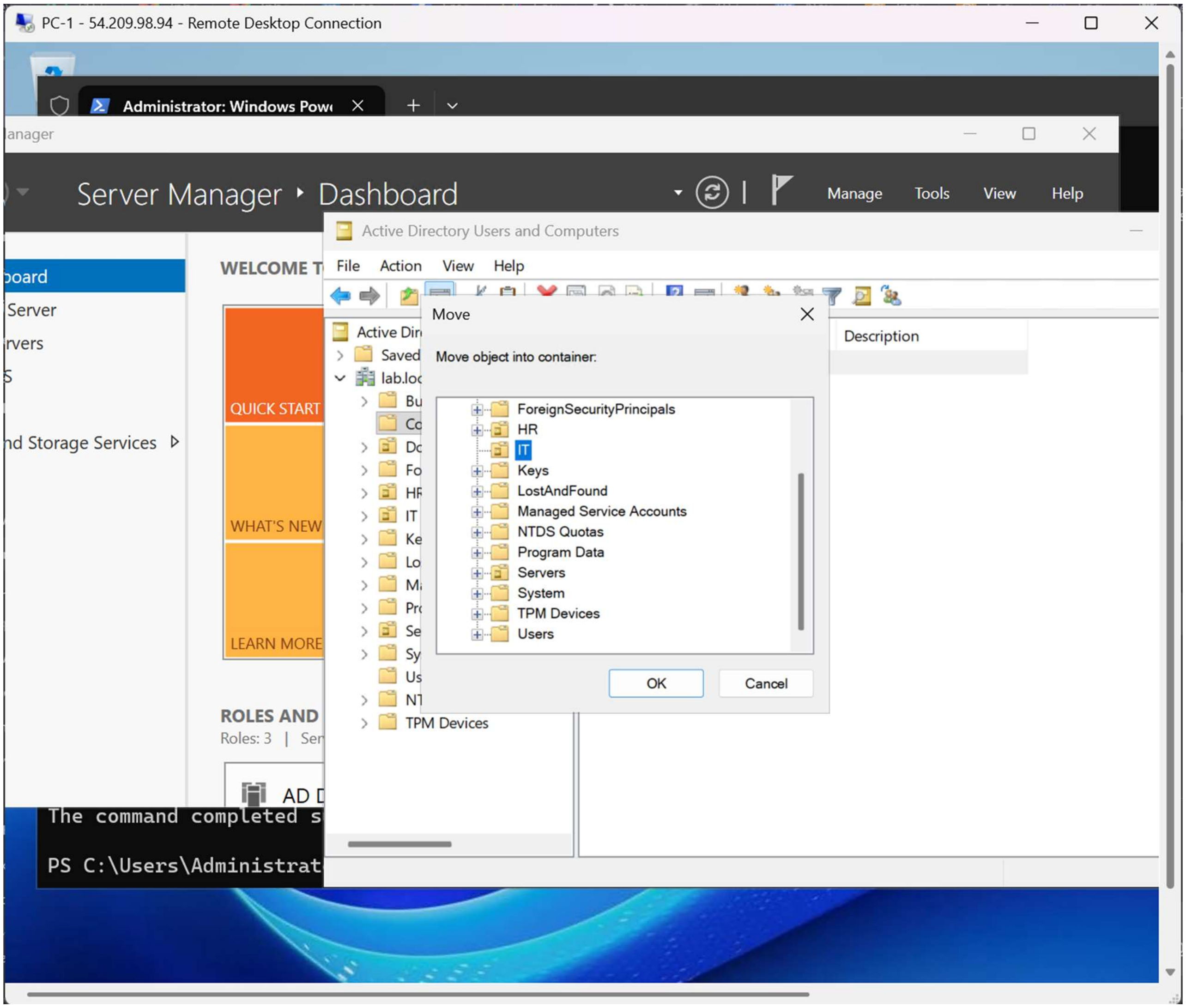
Step 12 – Move the Computer Object in Domain Controller

On the Domain Controller, open Active Directory Users and Computers, right-clicked the computer object and selected Move.



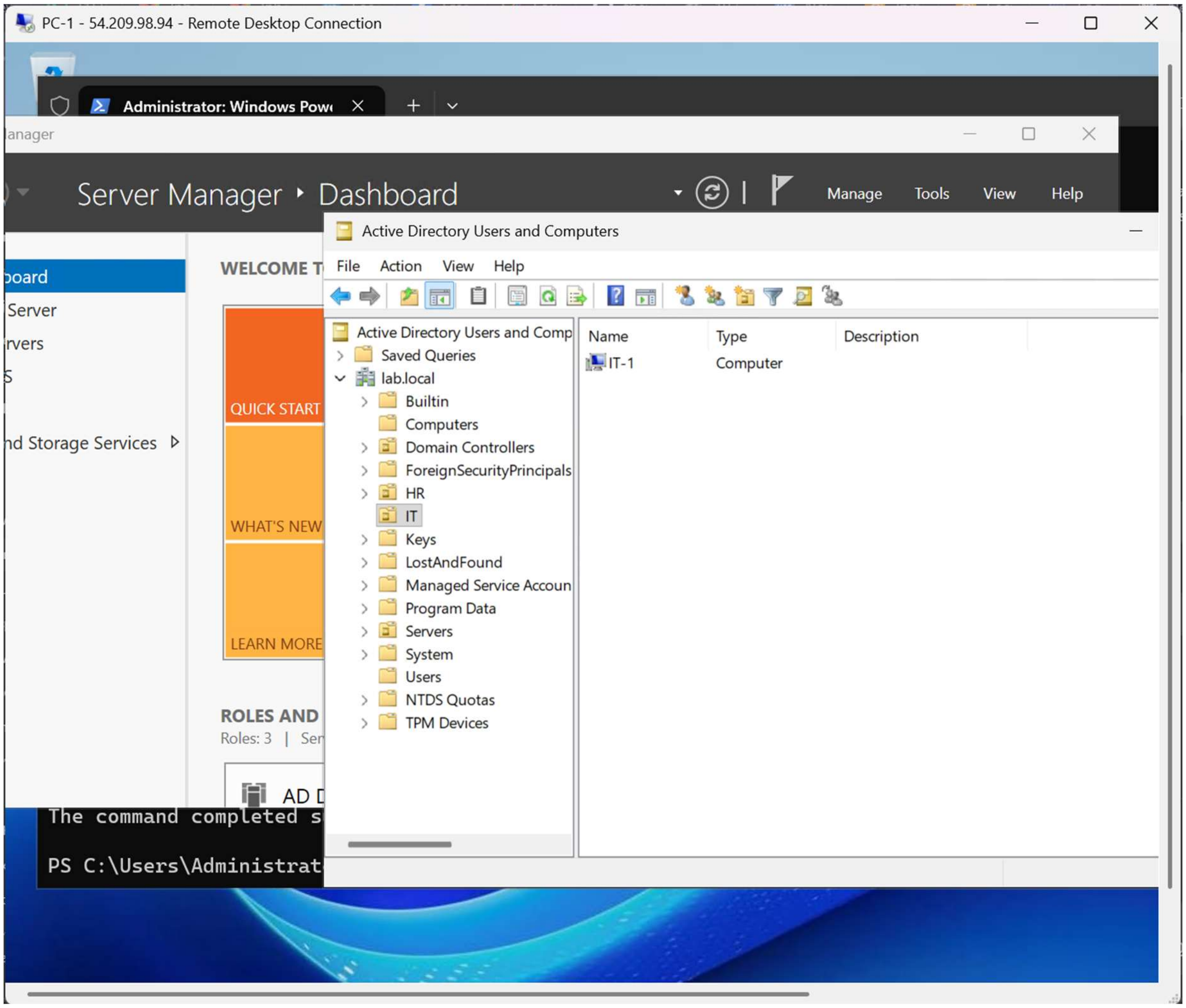
Step 13 – Select the Target OU

Selected the IT Organizational Unit as the destination for the computer account.



Step 14 – Verify OU Placement

Confirmed that the renamed computer (IT-1) appeared successfully inside the IT Organizational Unit.



Commands Used

- ping 10.0.0.8
- Test-NetConnection 10.0.0.8 -Port 53
- ipconfig /all
- Set-DnsClientServerAddress -InterfaceAlias Ethernet -ServerAddresses 10.0.0.8
- ipconfig /flushdns
- nslookup lab.local
- nltest /dsgetdc:lab.local

Conclusion

The client machine was successfully configured to use the Domain Controller as its DNS server, joined to the lab.local Active Directory domain, renamed to IT-1, moved into the IT Organizational Unit, and verified through DNS lookup and Active Directory management tools.